

AHMED SHA'BAN

MSc Candidate in Smart Systems Engineering | Machine Learning, Computer Vision and Multimodal AI

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RESEARCH PROFILE

Master's candidate in Smart Systems Engineering at An-Najah National University with a GPA of 3.58/4.00 and all 30 coursework credits completed. Hands-on research and engineering experience in machine learning, computer vision, multimodal AI, retrieval-augmented generation, efficient language-model deployment, and hardware-integrated smart systems. Developed and processed a 100,000-image research dataset and implemented AI pipelines using Qwen2.5, PyTorch, Hugging Face Transformers, FAISS, CUDA, and quantized models. Seeking a funded research internship, visiting-student position, or thesis research opportunity in AI and smart systems.

EDUCATION

MSc Candidate in Smart Systems Engineering

Sep 2022 - Present

An-Najah National University, Nablus, Palestine

GPA: 3.58/4.00; completed all 30 coursework credit hours and currently at the master's thesis research stage.

Research focus: multimodal AI, computer vision, retrieval-augmented generation, generative AI, and efficient model deployment.

BSc in Computer Science

Sep 2017 - Feb 2022

An-Najah National University, Nablus, Palestine

Foundation in software development, algorithms, databases, computer networks, system design, and intelligent applications.

RESEARCH EXPERIENCE

Master's Thesis Researcher

2025 - Present

An-Najah National University | Supervisor: Dr. Amjad Hawash

Developed and processed a large-scale dataset of 100,000 real images for multimodal image understanding and text-generation research.

Designed a retrieval-augmented generation workflow using FAISS for semantic retrieval and context-aware prompt construction.

Implemented experiments with Qwen2.5 language models, quantized GGUF models, PyTorch, Hugging Face Transformers, and CUDA-based GPU inference.

Built pipelines for data preparation, retrieval, prompt generation, generated-output management, and model evaluation.

Explored vision-language models, LoRA/QLoRA adaptation, efficient inference, and deployment on consumer GPU hardware.

SELECTED RESEARCH AND ENGINEERING PROJECTS

Multimodal Image-to-Story Generation System | Qwen2.5, FAISS, PyTorch, Transformers, CUDA

Built a multimodal RAG pipeline combining image understanding, semantic retrieval, structured prompt construction, and language-model generation for single-image and sequential-image scenarios.

Deep Learning Classification of E-Commerce Product Image Quality | Python, PyTorch, CNNs, Model Evaluation

Developed and evaluated CNN-based models for classifying product-image quality across five classes, using accuracy, precision, recall, F1-score, and confusion-matrix analysis.

Computer Vision-Based Autonomous Car Prototype | Computer Vision, OpenCV, Sensors, Real-Time Control

Built a prototype capable of detecting traffic signs and responding through real-time hardware control, integrating computer vision with embedded decision logic.

SELECTED PROJECTS - CONTINUED

Gold Price Decision Support System | *scikit-learn, Time Series, Feature Engineering*

Developed a machine-learning workflow for XAUUSD prediction using time-series features and compared Logistic Regression, Random Forest, and Support Vector Machine models.

Biometric Attendance and Management Integration | *C#, MySQL, Biometric Devices*

Integrated fingerprint devices with desktop applications and databases, handling real-time device communication, biometric data processing, and system workflows.

Swarm Robotics Coordination Prototype | *Python, Computer Vision, WebSockets, Jetson Nano*

Contributed to a four-robot JetBot system in which one robot becomes a leader based on visual detection and other robots coordinate through a server-mediated communication workflow.

PROFESSIONAL EXPERIENCE

Hardware and Technical Support Specialist

Jan 2020 - Jan 2025

Easy Life Technology, Nablus, Palestine

Assembled, diagnosed, maintained, and optimized desktop and laptop computer systems.

Troubleshoot hardware, operating-system, driver, and performance issues and provided practical customer solutions.

Developed strong hands-on knowledge of computer components, system performance, and hardware-software integration.

ADDITIONAL SOFTWARE AND SYSTEMS DEVELOPMENT

Developed PHP and MySQL management systems for point-of-sale operations, products, composite items, invoicing, mixed payments, reporting, and data export.

Designed an ASP.NET Core MVC gym-management architecture covering members, subscriptions, payments, attendance, cash handling, and audit trails.

Implemented database-backed desktop integrations and operational tools that connect business workflows with hardware devices and real-time data.

TECHNICAL SKILLS

Programming: Python, C++, C#, PHP, SQL, JavaScript

Machine Learning and AI: PyTorch, Hugging Face Transformers, scikit-learn, OpenCV, CNNs, deep learning, computer vision, model evaluation

Generative and Multimodal AI: Qwen2.5, vision-language models, retrieval-augmented generation, FAISS, prompt engineering, GGUF models, LoRA/QLoRA

Systems and Hardware: CUDA, GPU inference, IoT, sensors, biometric devices, embedded and real-time systems, hardware-software integration

Data and Development Tools: Data preparation, feature engineering, MySQL, Git, VS Code, Windows, Linux

LANGUAGES AND RESEARCH INTERESTS

Languages: Arabic (Native); English (B2 - Upper-Intermediate)

Research Interests: Efficient AI, multimodal systems, computer vision, generative AI, robotics, IoT, and trustworthy machine learning